

## Network Flow

- Maximum Flow: Problem Formulation
- Maximum Flow: Upper Bound
- Maximum Flow: Adding flow along paths
- Residual Network and Augmenting Path
- Ford-Fulkerson Algorithm – Max-Flow-Min-Cut Theorem
- Edmond-Karp Algorithm
- Maximum Flow: Variants and Applications

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## Max Flow: Problem Formulation

**Input:** A flow network  $G = (V, E, c)$ ,  $c : E \rightarrow \mathbb{R}^+$

$s \in V$  a source and  $t \in V$  a sink

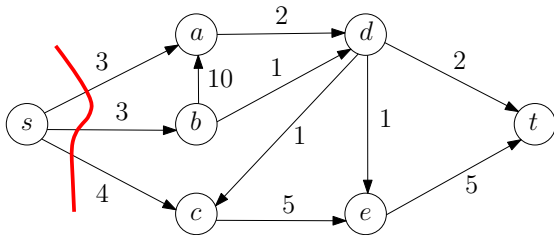
$f : E \rightarrow \mathbb{R}^+$  ( $f_e = f(e)$ ) is a flow if it satisfies

- 1 **capacity constraints**  $\forall e \in E : 0 \leq f_e \leq c_e$
- 2 **flow conservation constraints**  $\forall v \in V, v \neq s, t \quad f^{out}(v) = f^{in}(v)$

$$size(f) = f^{out}(s) = f^{in}(t)$$

**Output:** A flow  $f$  of maximum size

## Max Flow: Upper Bound

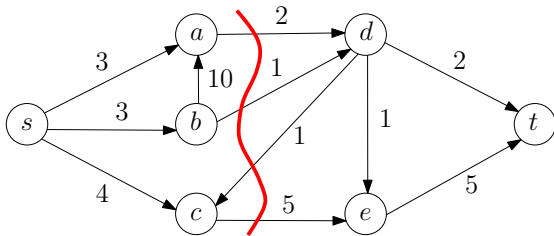


Consider the cut  $[\{s\}, \overline{\{s\}}]$

Any flow generated from  $s$  has to go through one of the cut edges

Hence, no flow can be of size bigger than  $3 + 3 + 4 = 10$

## Max Flow: Upper Bound



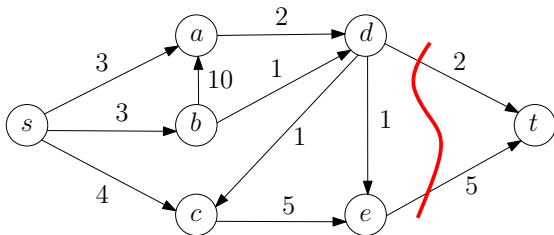
The same is true for any cut, Consider the cut  $[\{s, a, b, c\}, \{d, e, t\}]$

Any flow generated from  $s$  has to go through one of the cut edges

Hence, no flow can be of size bigger than  $2 + 1 + 5 = 8$

This is a tighter bound than the one we got from  $[\{s\}, \overline{\{s\}}]$

## Max Flow: Upper Bound



The same is true for any cut, Consider the cut  $[\overline{\{t\}}, \{t\}]$

Any flow generated from  $s$  has to go through one of the cut edges

Hence, no flow can be of size bigger than  $2 + 5 = 7$

This is a tighter bound than the one we got from  $[\{s, a, b, c\}, \{d, e, t\}]$

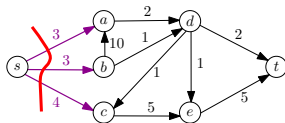
## Max Flow: Upper Bound

$s - t$  cut

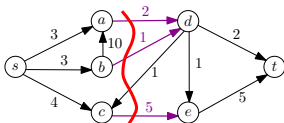
$A \subset V$ , an  $s - t$  cut,  $[A, \bar{A}]$ , is a cut in  $G$  with  $s \in A$  and  $t \in \bar{A}$

Capacity of an  $s - t$  cut: sum of capacities of edges going from  $A$  to  $\bar{A}$

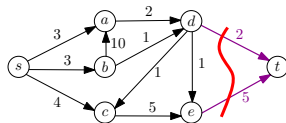
$$c([A, \bar{A}]) = \sum_{e \text{ outgoing from } A} c_e$$



An  $s - t$  cut of capacity 10



An  $s - t$  cut of capacity 8



An  $s - t$  cut of capacity 7

## Max Flow: Upper Bound

Let  $f$  be a flow in  $G$  and let  $[A, \bar{A}]$  be any  $s - t$  cut in  $G$ , then

$$\text{size}(f) \leq c([A, \bar{A}])$$

**Proof:** Let  $[A, \bar{A}]$  be any  $s - t$  cut. By definition we know that

$$\text{size}(f) = f^{\text{out}}(s) = f^{\text{out}}(s) + \sum_{s \neq v \in A} (f^{\text{out}}(v) - f^{\text{in}}(v)) \quad \text{just adding 0's}$$

$$= f^{\text{out}}(s) + \sum_{s \neq v \in A} f^{\text{out}}(v) - \sum_{s \neq v \in A} f^{\text{in}}(v)$$

$$= f^{\text{out}}(A) - f^{\text{in}}(A)$$

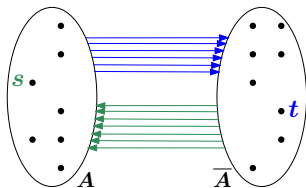
$$= \sum_{e \text{ outgoing from } A} f_e - \sum_{e \text{ incoming to } A} f_e \quad \text{flows on other edges cancel}$$

$$\leq \sum_{e \text{ outgoing from } A} c_e - \sum_{e \text{ incoming to } A} f_e \leq \sum_{e \text{ outgoing from } A} c_e = c([A, \bar{A}])$$

## Max Flow: Upper Bound

Let  $f$  be a flow in  $G$  and let  $[A, \bar{A}]$  be any  $s - t$  cut in  $G$ , then

$$\text{size}(f) \leq c([A, \bar{A}])$$



Tightest upper bound will come from a  $s - t$  cut of minimum capacity

$[A^*, \bar{A}^*]$  be an  $s - t$  cut with minimum capacity

▷ min- $s - t$ -cut

We get the corollary

$$\text{size}(f) \leq c([A^*, \bar{A}^*])$$